Aryan Naik

aryannaik1205@gmail.com | github.com/RobotsAndAryan | UCL MEng Robotics and Artificial Intelligence

EDUCATION

University College London — MEng Robotics and Artificial Intelligence (2nd Year, Expected 2029)

PROJECTS

Aerospace Avionics Firmware — Student Rocketry Telemetry Payload

- Delivered a flight-ready avionics payload capable of autonomous parachute deployment by implementing a finite state machine in C++ across a dual-MCU embedded architecture, achieving reliable safety-critical descent control under live flight conditions
- Enabled continuous inertial state monitoring across the full flight envelope by integrating IMU, altimeter, and accelerometer sensors over I2C, producing validated attitude estimation and altitude logging throughout ascent and descent
- Sustained uninterrupted telemetry and onboard data retention by designing wireless transmission and flash logging subsystems in parallel, ensuring no data loss across the flight cycle
- Maintained hardware-software co-design integrity across the avionics stack by applying power budgeting, signal conditioning, and UART/SPI serial communication protocols, meeting all signal threshold requirements verified during bench testing prior to launch

Full-System Robotics Integration — Competition Platform

- Delivered a complete, competition-ready robotic platform that passed all acceptance testing criteria by integrating microcontrollers, motor drivers, sensors, and power electronics from schematic to physical assembly within a fixed competition deadline
- Restored full closed-loop sensor-actuator control following hardware faults by applying systematic fault-finding and signal integrity verification across the breadboard circuit, returning the platform to operational status within the build window
- Reduced mechanical iteration time by producing structural components, enclosures, and mounting geometry in Autodesk Fusion 360 with defined tolerances, enabling direct handoff to 3D printing without rework

TECHNICAL SKILLS

Embedded and Firmware: C++, real-time firmware, interrupt-driven execution, finite state machine design, UART, SPI, I2C, sensor fusion, inertial sensing and state estimation, power budgeting, signal conditioning

Electronics and Hardware: Breadboard circuit design, component selection, voltage regulation, H-bridge and servo motor control, ultrasonic and IR distance sensors, hardware debugging, systems integration and acceptance testing

Mechanical and CAD: Autodesk Fusion 360, structural design, tolerancing, rapid prototyping, 3D printing

Domain: Safety-critical embedded software, sounding rocket avionics, autonomous recovery systems, defence electronics, aerospace engineering, mechatronics

ADDITIONAL

Disciplines of interest aligned to Babcock core sectors: naval systems, mechanical engineering, electrical engineering, and aerospace. Familiar with the operational demands of sustaining critical assets in high-reliability environments through safety-critical firmware development and systems integration work. Keen to apply embedded and systems engineering skills within a defence and naval engineering context, including asset maintenance and long-term operational reliability.

Aryan Naik aryannaik1205@gmail.com github.com/RobotsAndAryan

Hiring Team Babcock International Group

Dear Hiring Manager,

I am writing to apply for the Engineering Undergraduate Summer Internship at Babcock. What draws me specifically to Babcock is its reputation for sustaining complex, safety-critical infrastructure on behalf of national armed forces and governments. The combination of deep specialist engineering with long-term asset management in high-reliability environments is precisely the kind of technically demanding context in which I want to develop as an engineer. As a second-year MEng Robotics and Artificial Intelligence student at University College London, I am keen to apply my embedded systems and hardware engineering experience within a defence and aerospace setting of genuine operational consequence.

My most directly relevant project is the development of aerospace avionics firmware for a student rocketry telemetry payload. Working on a dual-MCU embedded architecture, I wrote interrupt-driven C++ firmware integrating IMU, altitude, and acceleration sensors over I2C to achieve real-time attitude estimation and flight data logging. I implemented a finite state machine governing autonomous parachute deployment under safety-critical flight conditions, alongside wireless telemetry transmission and onboard flash logging. This project required me to balance hardware-software co-design, power budgeting, and signal integrity under strict reliability constraints, producing a functional flight computer capable of autonomous recovery decisions in a live launch environment.

I would bring to Babcock a practical grounding in embedded firmware, sensor integration, and safety-critical system design, alongside the rigour developed through working on real hardware under competition and launch conditions. I am eager to learn how engineering decisions are made and validated at the scale and accountability that defence programmes demand, and to understand how professional teams manage complexity, certification, and operational risk across long-lifecycle systems.

Yours sincerely, Aryan Naik